Solutions

Question 1 on Wind and Photovoltaic Power

a)

i) Rated wind speed

$$P_w = \frac{1}{2}\rho A v_w^3 C_{p-max}$$

$$\Rightarrow v_r = \left(\frac{2P_{rated}}{\rho\pi R^2 C_{p-max}}\right)^{\frac{1}{3}} = 11.34 \text{ m/s}$$

[0.5 mark each]

ii)

Maximum power generated at wind speed of 6 m/s

$$P_{w6} = \frac{1}{2}\rho A 6^3 C_{p-max} = 1.4 \text{ MW}$$

[0.5 mark]

Maximum power generated at wind speed of 12 m/s

$$P_{w12} = \frac{1}{2}\rho A 12^3 C_{p-max} = 11.25 \text{ MW}$$

But this is above the rated power of the wind turbine.
[1 mark]

So the maximum power generated at 12 m/s is the rated power 9.5 MW

[0.5 mark]

iii)

Tip-speed ratio is given by:

$$\lambda = \frac{\omega_T \times R}{v_w} \Rightarrow \omega_{T-opt} = \frac{\lambda_{opt} \times v_w}{R}$$

[1 mark]

Variable speed operation is required between the cut-in and

$$\left(\frac{\lambda_{opt} \times v_{w-cutin}}{R} \times \frac{60}{2\pi}\right) \leq \omega_{T-opt} \leq \left(\frac{\lambda_{opt} \times v_r}{R} \times \frac{60}{2\pi}\right)$$

$$\Rightarrow 2.86 \text{ rpm} \leq \omega_{T-opt} \leq 10.83 \text{ rpm}$$

Model Answers

[2 marks]

iv)

Beyond the rated wind speed, the rotational speed of the turbine is kept constant by pitching the turbine blades.

[1 mark]

Rotational speed of the turbine at 12 m/s wind speed

$$\omega_T = \frac{\lambda_{opt} \times v_r}{R} \times \frac{60}{2\pi} = 10.83 \text{ rpm}$$

[1 mark]

v)

Synchronous speed (in rpm)

$$N_s = \frac{60 \times f}{p/2} \Rightarrow p = \frac{120 \times f}{N_s}$$

[0.5 mark]

For

$$f \geq 8 \text{ Hz}, p \geq \frac{120 \times f}{N_{s-min}} = 335.1$$

[1 mark]

Minimum number of poles needed is 336

[0.5 mark]

b)

i)

Irradiance under STC is 1000 W/m^2

[1 mark]

Area of each module

$$A = \frac{P_{\text{rated}}}{1000 \times \eta} = 1.83 \text{ m}^2$$

[1 mark]

ii)

$$P_{\text{arr}} = N_{\text{mod}} \times 600 \times A \times \eta \times 10^{-3} = 99 \text{ kW}$$

Model Answers

[1 mark]

iii)

The current (i.e., power) fed by the 3 affected strings would be zero as the corresponding string diodes are reverse biased. This is true whether one or more than one module within a string is shaded.

[1 mark]

$$P_{\text{out}} = N_{\text{mod}} \times P_{\text{rated}} \times \frac{N_{str} - 3}{N_{str}} = 115.5 \text{ kW}$$

[1 mark]

iv)

The current (i.e., power) fed by the affected string would be zero as the corresponding string diode is reverse biased.

$$P_{\text{out}} = N_{\text{mod}} \times P_{\text{rated}} \times \frac{N_{str} - 1}{N_{str}} = 148.5 \text{ kW}$$

[1 mark]

v)

- For a fixed tilt angle, maximum irradiance over the course of a year is usually obtained when the tilt angle is equal to the latitude. Hence the tilt angle of the panels should be 40° facing southward.
- The tilt angle could be larger than the latitude if higher fraction of direct light is incident during the winter period, while lower title angles could be used for a higher fraction of direct light during the summer months.

[1 mark each]

i)

On the summer solstice day, the declination angle $\delta = 23.45°$
Optimum tilt angle $\beta = \phi - \delta = 16.5°$

[1 mark]

On the winter solstice day, the declination angle $\delta = -23.45°$
Optimum tilt angle $\beta = \phi - \delta = 63.5°$

[1 mark]

Model Answers

Question 2 on balancing, Voltage Control and Security of Supply

Part A

a)

Statements along the following lines would be sufficient for the full mark:

- Part-loaded generation, including
  - Enhancing flexibility of nuclear generation
- Pumped-storage hydro generation
- Flexibility
- Standby generation (standing reserve)

[1 mark]

b)

Statements along the following lines would be sufficient for the full marks:

Advantages

- Potentially less expensive solution
- Increased efficiency of generation that is not anymore part loaded
- Potentially less curtailment of renewables
- Can provide standing reserve without cost increase

[maximum 1.5 marks; subtract 0.5 marks for each not mentioning point above up to total 1.5 marks; add 0.5 for any relevant point not mentioned above]

Disadvantages

- More actors involved increasing complexity and adds need for coordination
- Load recovery for some of technologies
- Might need contracting more capacity due to potentially higher uncertainty of some of technologies

[maximum 1.5 marks; subtract 0.5 marks for each not mentioning point above; add 0.5 for any relevant point not mentioned above]

Part B

a)

Fit and forget approach: limit distributed generation capacity to level

[0.5 mark]

Generation curtailment: during critical conditions distributed generation is curtailed to keep network constraints within limits.

[0.5 mark]

Reactive power compensation: reactive compensator at distributed generation connection point could be set or controlled in a such way to reduce the impact of voltage rise effect.

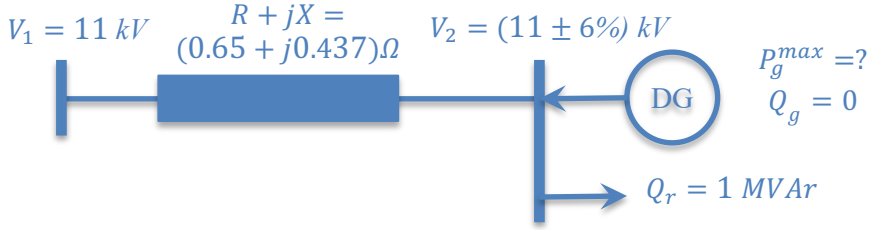
[0.5 mark]

Voltage control: different voltage control techniques (setting different tap changer position/target voltage, adding in-line voltage regulators and area-based voltage control) could be applied to keep voltage withing limits.

[0.5 mark]

Model Answers

b)



[1 mark; Qg does not need to be shown above for the full mark]

$$P_g^{max} = \frac{(V_2^{max} - V_1)V_2^{max}}{R} + \frac{Q_r \cdot X}{R}$$

$$P_g^{max} = \frac{(1.06 \cdot 11 - 11) \cdot 1.06 \cdot 11}{0.65} + \frac{1 \cdot 0.437}{0.65}$$

$$P_g^{max} = 12.5MW$$

[1 mark]

Part C

a)

Capacity outage probability table for the generation plant is

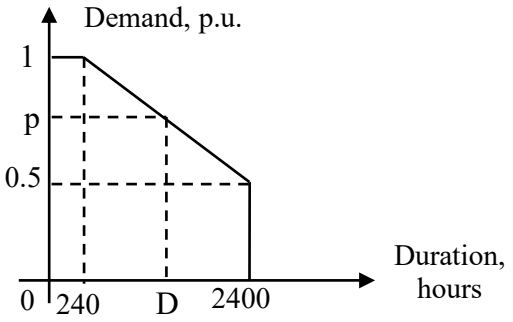
Capacity in service, MW	Probability
17	0.85 x 0.75 x 0.65 = 0.4144
15	0.85 x 0.75 x (1-0.65) = 0.2231
12	0.85 x (1-0.75) x 0.65 = 0.1381
10	0.85 x (1-0.75) x (1-0.65) = 0.0744
7	(1-0.85) x 0.75 x (1-0.65) = 0.0731
5	(1-0.85) x 0.75 x (1-0.65) = 0.0394
2	(1-0.85) x (1-0.75) x 0.65 = 0.0244
0	(1-0.85) x (1-0.75) x (1-0.65) = 0.0131

The total probability for all eight possible capacity states is 1.

[4 marks; subtract 1 mark for any missing capacity in service and wrong probability up to a maximum of subtracted 4 marks]

Model Answers

b)
Energy not supplied is calculated from the LDC in the following way



For a given $0.5 \leq p < 1$, D could be calculated from linear equation:

$$D = 240 + \frac{2400 - 240}{0.5 - 1}(p - 1) = 4560 - 4320 \cdot p$$

And energy not supplied, ENS, is:

$$ENS = (1 - p) \cdot C \cdot \big(240 + 0.5 \cdot (D - 240)\big) = (1 - p) \cdot C \cdot (120 + 0.5 \cdot D)$$

Where total generation capacity, C, is 17 MW.

For a given $p \leq 0.5$, ENS could be calculated as

$$ENS = (1 - 0.5) \cdot C \cdot \big(240 + 0.5 \cdot (2400 - 240)\big) + (0.5 - p) \cdot C \cdot 2400$$
$$= (1860 - p \cdot 2400) \cdot C$$

Expected energy not supplied is equal to $ENS \cdot Probability$

Capacity, MW	Probability	p	D, hours	ENS, MWh	EENS, MWh
17	41.44%	1	0	0	0
15	22.31%	0.8824	748.24	988.24	220.50
12	13.81%	0.7059	1510.6	4376.5	604.50
10	7.44%	0.5882	2018.8	7905.9	588.00
7	7.31%	0.4118		14820	1083.7
5	3.94%	0.2941		19620	772.54
2	2.44%	0.1176		26820	653.74
0	1.31%	0		31620	415.01
Total	**100.0%**				**4338**

Total EENS is 4338 MWh.

[5 marks; subtract 2 marks if it not clearly explained how EENS is calculated; subtract 1 mark for any wrong pair ENS/EENS, including total EENS up to a maximum of subtracted 5 marks]

Model Answers

c)
Energy not supplied when $p = 0.5$ is

$$ENS_{0.5} = (1 - 0.5) \cdot 17 \cdot (240 + 0.5 \cdot (2400 - 240)) = 11{,}220MWh$$

Given the equivalent firm capacity source never fails, probability is 1 and hence firm capacity is greater than 0.5 p.u. Alternatively ENS of 12 MW is 4376.5 MWh that is greater than total EENS meaning firm capacity is greater than 12 MW. ENS (or EENS) of the firm capacity is equal to the total EENS.

$$ENS = (1 - p) \cdot C \cdot (120 + 0.5 \cdot D) = (1 - p) \cdot C \cdot \left(120 + 0.5 \cdot (4560 - 4320 \cdot p)\right)$$
$$= C \cdot (2160 \cdot p^2 - 4560 \cdot p + 2400)$$

$$p = \frac{4560 \pm \sqrt{4560^2 - 4 * 2160 * \left(2400 - \frac{ENS}{C}\right)}}{2 \cdot 2160}$$

$$p = 0.7074$$

Contribution is 71% and in MW is $0.7074 \cdot 17 = 12.026MW$. Firm capacity is 12.026MW.

[3 marks; subtract 2 marks if it is not clearly explained how p *or* $p \cdot C$ is calculated; subtract 1 mark for wrong firm capacity]

Model Answers

Question 3 on Low Carbon Heat

a)

The purpose of the CCC is to:

- advise UK Government and Devolved Administrations on emissions targets and report to Parliament on progress made in reducing greenhouse gas emissions and preparing for climate change.
- provide independent advice on setting and meeting carbon budgets and preparing for climate change.
- monitor progress in reducing emissions and achieving carbon budgets and targets.
- conduct independent analysis into climate change science, economics and policy.
- engage with a wide range of organisations and individuals to share evidence and analysis.

[2 marks]

The reason for the report is because the CCC must report to Parliament its assessment of the Government's performance in combatting climate change and measure its actions against its national and international legal obligations to which the UK is committed.

The key messages include:

- UK Government now has a solid Net Zero strategy.
- Policy gaps in a number of areas remain, e.g. land use, building energy efficiency.
- Tangible progress in a wide range of areas lags policy ambition, e.g. building energy efficiency

[2 marks]

b)

This is because there are sectors of the economy described as "hard-to-treat" where at present there are limited options available to reduce GHG emissions. For example, aviation and agriculture.
[1 mark]

BECCS converts biomass into energy. It is regarded as having "net neutral" CO_2 emissions as the biomass is only releasing the CO_2 absorbed when it was a crop. The CO_2 emissions are captured using a chemical process and then transported to permanent storage in a geological reservoir. Hence overall the emissions are regarded as negative.

DACCS uses chemical processes to remove CO2 from the atmosphere. The CO2 can then be stored permanently in a geological reservoir. Its main drawback is that at present it is very energy intensive.
[2 marks]

Model Answers

c)

ASHP transfers thermal energy from the outside air to the area to be heated. There are 4 stages to the cycle and these comprise:

- Evaporation - The refrigerant passes through a heat exchanger where it evaporates absorbing heat in the process.
- Compression - The heated refrigerant is then compressed to a higher pressure using an electrically driven compressor.
- Condensation - The compressed refrigerant passes through another heat exchanger located in the space to be heated and condenses, changing state back to a liquid and rejecting heat in the process.
- Expansion – The liquid refrigerant passes through an expansion device lowering its pressure prior to returning to the evaporator.

[2 marks]

This operates on the same principle as an ASHP except it incorporates an additional heating source such as a condensing gas boiler to provide further heat capacity for both space and water heating.
[1 mark]

Benefits:

- A hybrid heat pump may permit a smaller size heat pump to be installed thereby reducing electricity demand and in particular peak demand. This might avoid the need to reinforce the electricity network and reduce the amount of generation capacity required.
- A hybrid heat pump using a condensing combination gas boiler can produce hot water on demand thereby reducing the need for hot water storage.
- A hybrid heat pump with its higher heat output is likely to be better suited to buildings with poor insulation.

Drawbacks

- A hybrid heat pump will have higher CO2 emissions if the gas boiler is supplied by natural gas. This can be reduced or avoided if biogas or hydrogen is used.
- A hybrid heat pump is likely to cost more due to the additional heat source.

[2 marks]

A heat pump uses a “vapour compression cycle” to absorb heat from an external source, e.g. air, water, ground. This exceeds the energy consumption of heat pump compressor thereby leading to an efficiency exceeding 100%.
[1 mark]

d)
i)

- Building internal set temperature, K
- Building external temperature, K

Model Answers

- Other weather factors, e.g. solar gain, wind, K
- Fabric (walls, windows, etc) thermal transmittance, Wth/m2K
- Air changes or permeability, m3/m2.h @ 50Pa specified pressure
- Other heat sources, e.g. occupants, appliances, etc

[1 mark]
4 are required for the full mark

e)
i)
Typical annual building space heat demand (Whth)

= (Tint – T ext) * Average thermal transmittance * Building floor area*Nos of days heating is on*Nos of hours heating is on during the day

= (18-5.5) *2.0*120*100*10 = 3,000,000 Whth or 3.0 MWhth

Where:
Tint = Building internal set temperature, °C
Text= Building external set temperature, °C

Assumptions:
Estimate of heat demand does not take account of:

- other weather factors such as wind, solar gain.
- building air changes
- other heat sources

[3 marks]

ii)
Peak heat demand (W) = (Tint – T ext) * Average thermal transmittance * Building floor area
= (18-(-7))*2.0*120 = 6000Wth or 6.0kWth
[2 marks]

iii)
The peak heat demand will also reduce by 20%, i.e. to 4.8 kWth giving a reduction in peak heat of 1.20kWth
[1 marks]

Model Answers

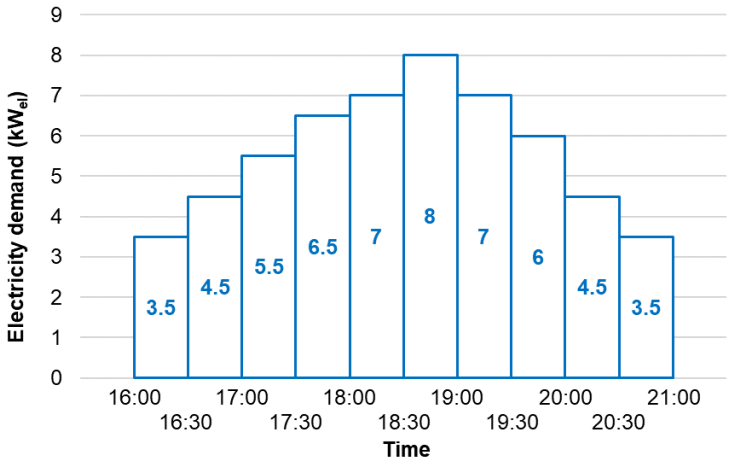
Question 4 on Flexibility, Hydro Power and Offshore Transmission

Demand Flexibility

a)
The application of DSR schemes tends to disturb the natural diversity of loads. DSR applications rely on the ability of controlled devices (appliances) to reschedule operation so DSR redistributes the load but does not necessarily reduce the total energy consumed by the device or appliance. Load reduction periods will be generally followed by demand payback or load recovery, the duration of which will depend on the interrupted process and the nature of the storage. Reducing demand from thermostatically controlled loads generally results in energy payback as a consequence of the physics of heat transfer from the building or appliance to the surrounding space and vice versa combined with the objective to follow the target temperature profile.
(1 mark for the explanation of payback effect)

Demand payback can affect the volume of service (e.g. peak reduction) delivered by DSR providers, as the demand during recovery period generally increases compared to before the intervention, and can potentially exceed the original peak demand that DSR was trying to minimise. It may therefore be necessary for the grid or system operators to contract for a higher volume of flexibility from DSR than nominally required in order to offset the effect of demand payback.
(1 mark for the explanation of how payback affects the volume of service)

b)
i)
Peak electricity demand for the household can be found by dividing the peak heat demand with the relevant COP value. In this example the peak electricity demand is equal to: $P_{peak}^{el} = 16 \div 2 = 8\ \mathrm{kW_{el}}$.
The household's electricity demand diagram for the period in question is given below:



(1 mark for the correct peak demand)
(1 mark for the correct chart)

ii)
To calculate the correct size of thermal energy storage (TES), it is necessary to establish both the requirements for thermal energy and for thermal power (i.e. heat discharge rate) from TES in order to limit the electrical peak to 5 $\mathrm{kW_{el}}$.

Model Answers

Equivalent electrical energy requirements can be found by identifying those half-hourly intervals when the electricity demand exceeds the specified limit of 5 kW_{el}, summing up the values by which this limit is exceeded, and multiplying the sum with the duration of the unit time interval (0.5 h):

$$E^{el} = 0.5 \cdot [(5.5 - 5) + (6.5 - 5) + (7 - 5) + (8 - 5) + (7 - 5) + (6 - 5)] = 5\ \text{kWh}_{\text{el}}$$

The associated heat energy storage requirements are found by multiplying this value by the COP:

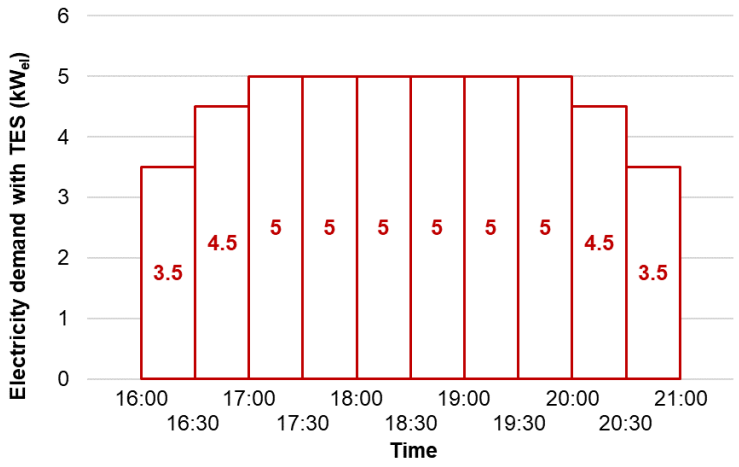
$$E^{th} = E^{el} \cdot COP = 5 \cdot 2 = 10\ \text{kWh}_{\text{th}}$$

Based on the fixed TES duration of 2 hours, this volume of energy would require a heat power rating of 10/2 = 5 kW_{th}.; however, it needs to be verified whether this level of heat power would be sufficient to limit the instantaneous electricity consumption to 5 kW_{el}.

Peak electricity demand occurs between 18:30 and 19:00 at the level of 8 kW_{el}. To reduce this by 3 kW_{el} to 5 kW_{el}, heat demand would need to reduce by 6 kW_{th}, given that COP = 2. This means that TES needs to have the heat power rating of at least **6 kW_{th}**, which further means that, due to the fixed energy-to-power ratio of 2, its size in terms of energy has to be at least **12 kWh_{th}**.

(1 mark for the correct heat energy requirement)
(1 mark for the correct heat power requirement)
(1 mark for the correct final TES size)

The resulting electricity demand diagram is given below:



(1 mark for the correct chart)

iii)
If COP is equal to 2.5, the electricity demand exceeds the 5 kW_{el} limit during 4 half-hourly intervals between 17:30 and 19:30. Similar to before, the heat energy required to reduce the electrical peak to the 5 kW_{el} limit is found as:

$$E^{th} = COP \cdot E^{el} = 2.5 \cdot 0.5 \cdot [(5.2 - 5) + (5.6 - 5) + (6.4 - 5) + (5.6 - 5)] = 3.5\ \text{kWh}_{\text{th}}$$

Answers

This energy rating would be equivalent to the heat power rating of 1.75 kW_{th}.

On the other hand, the heat power required in the highest demand interval is equal to $(16 - 5 \cdot 2.5) = \mathbf{3.5\ kW_{th}}$, which represents a more stringent requirement given the fixed energy-to-power ratio of 2. Therefore, the necessary heat storage size in energy terms is equal to **7 $\mathbf{kWh_{th}}$**.

(2 marks for the correct TES size)

Model Answers

Hydro Power

c)

i)

Effective head $h = h_{hw} - h_{tw} = 120 - 30 = 90$ m

[1 mark]

ii)

Rated power is given by $P_r = \rho g h Q \eta$

[1 mark]

The volume flow rate

$$Q = \frac{P}{\rho g h \eta} = \frac{15 \times 600 \times 10^6}{1000 \times 9.81 \times 90 \times 0.95} = 10{,}730 \text{ m}^3/s$$

[1 mark]

iii)

The rated speed of a synchronous generator (in rpm) is

$$N \text{ (in rpm)} = \frac{f \text{ (in Hz)} \times 60}{p/_2}$$

[1 mark]

$$\Rightarrow p = \frac{2 \times f \text{ (in Hz)} \times 60}{N \text{ (in rpm)}} = 60$$

[1 mark]

Offshore Transmission

d)

- Sub-sea cables have much larger capacitance per unit length compared to that of an equivalent overhead line.
- Charging current drawn by the cable capacitance increases with cable length (due to larger total capacitance) and operating voltage ($|i_c| = \omega C V$). For long high-voltage cables, charging current could be prohibitively high and limiting the active current carrying capacity of the cable.

Model Answers

- Use of HVDC virtually eliminates the above problem by drawing negligibly small charging current and thereby, retaining the current (and hence active power) transfer capability of high-voltage long-distance cables.

[1 mark each]

e)

- For an LCC HVDC link, power flow is reversed by flipping the direct voltage polarity at both ends as current direction through the HVDC link cannot be reversed. For a VSC HVDC link, direct voltage polarity is fixed, and power flow is reversed by reversing current flow through the HVDC link.

- Polymeric cables cannot withstand slow reversal of voltage polarity and hence can only be used with VSC HVDC but not LCC HVDC.

[1 mark each]

Model Answers